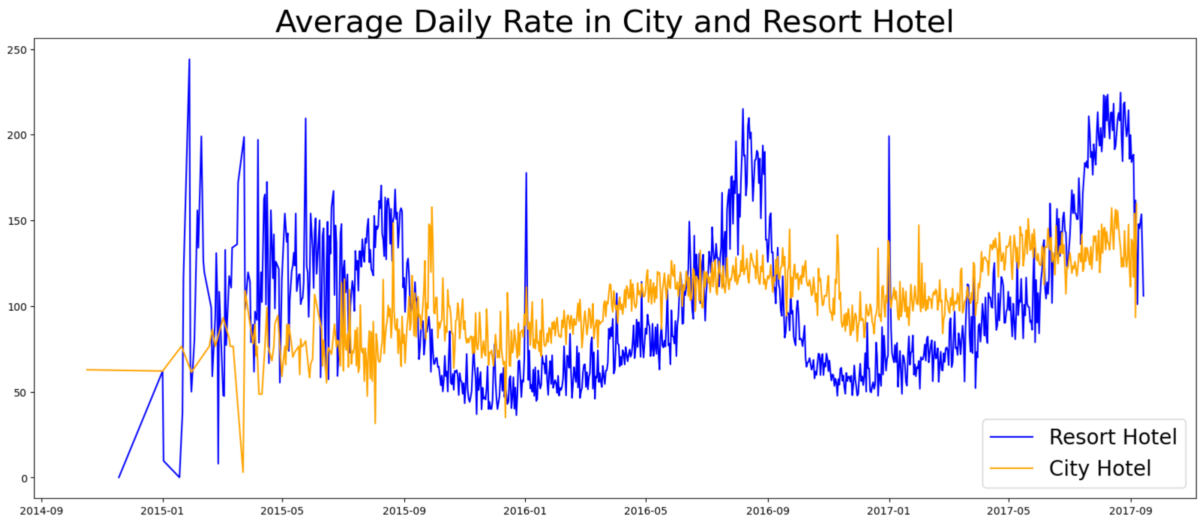
Hotel Booking Analysis Report: ADR Trends and Churn Insights (2016 Onwards)

Introduction

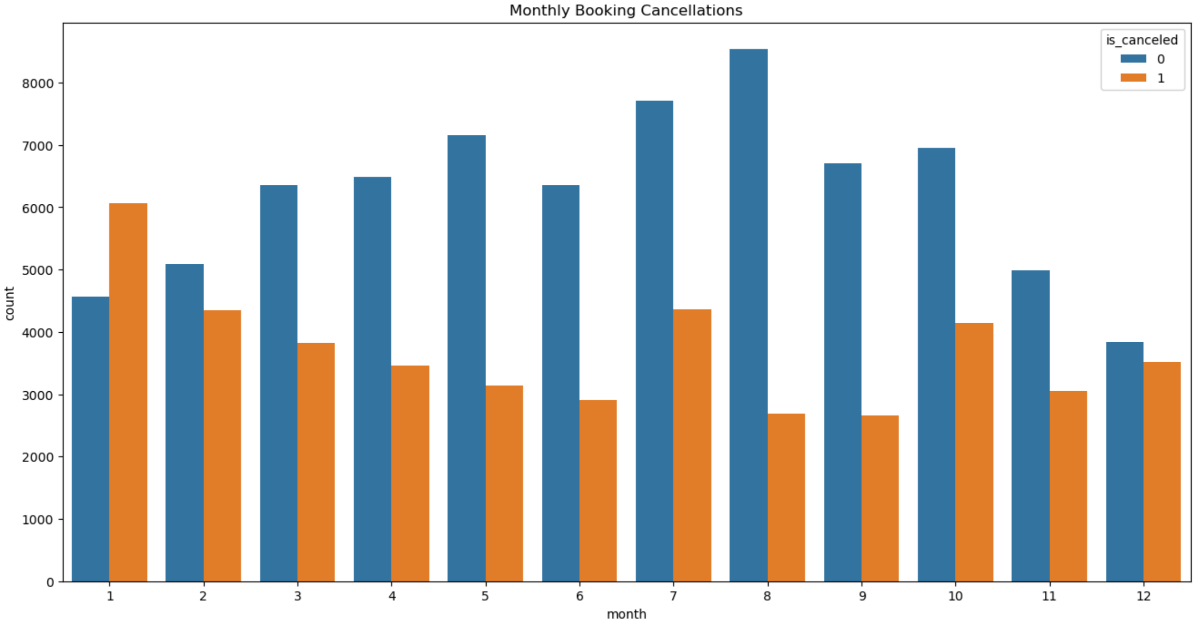
This report analyzes the **Average Daily Rate (ADR)** of hotel bookings from 2016 onwards and highlights cancellation patterns (churn). The purpose is to understand how customer behavior affects revenue and which patterns can help reduce churn and improve bookings.

Key Definitions

- **ADR (Average Daily Rate):** The average rental income per paid occupied room per day.



- **Churn (Cancellation):** In this context, churn refers to



bookings that were canceled (`is_canceled = 1`).

Data Summary

The data includes hotel reservations with the following features:

- `reservation_status_date` (date of reservation status)
- `adr` (average daily rate)
- `is_canceled` (whether the reservation was canceled or not)

Data Cleaning

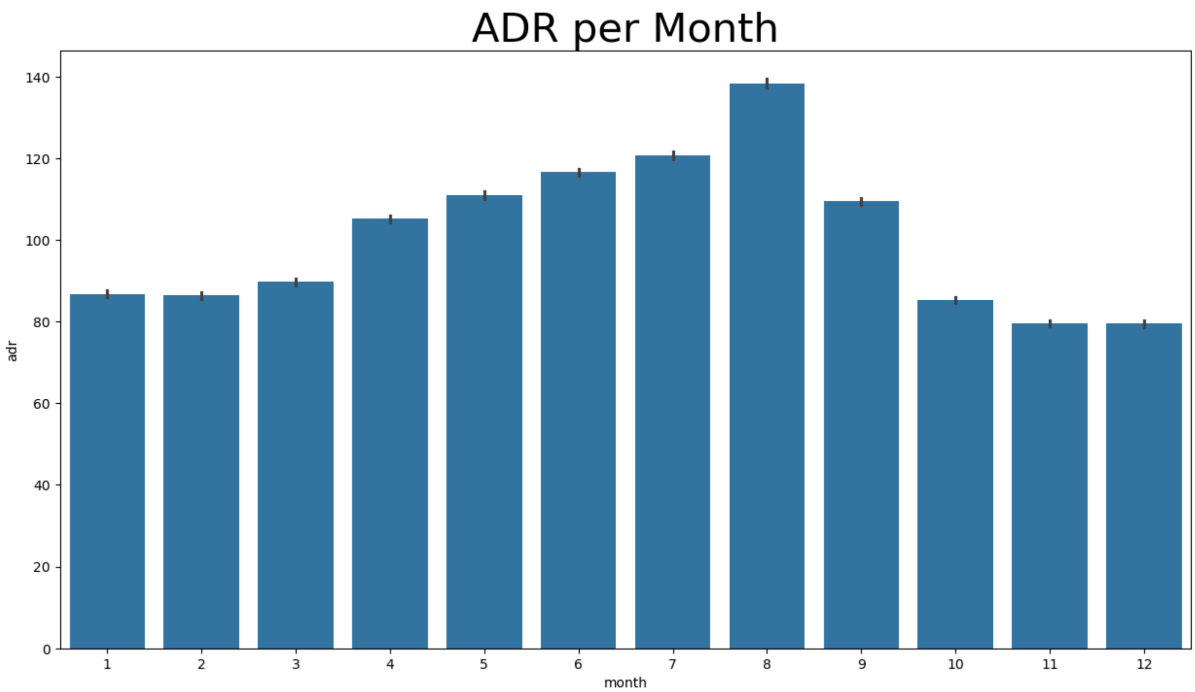
To ensure consistency:

- We filtered the data to include only records from **January 2016 onwards**.
- Removed entries with missing or invalid dates.
- Converted `reservation_status_date` to datetime format.

ADR Trend Analysis

Observations:

- Bookings that were **not canceled** (churn = 0) showed relatively **stable ADR trends**, with seasonal increases during peak travel months (e.g., summer and winter holidays).
- **Canceled bookings** had generally **higher ADRs**, indicating customers are more likely to cancel expensive bookings.

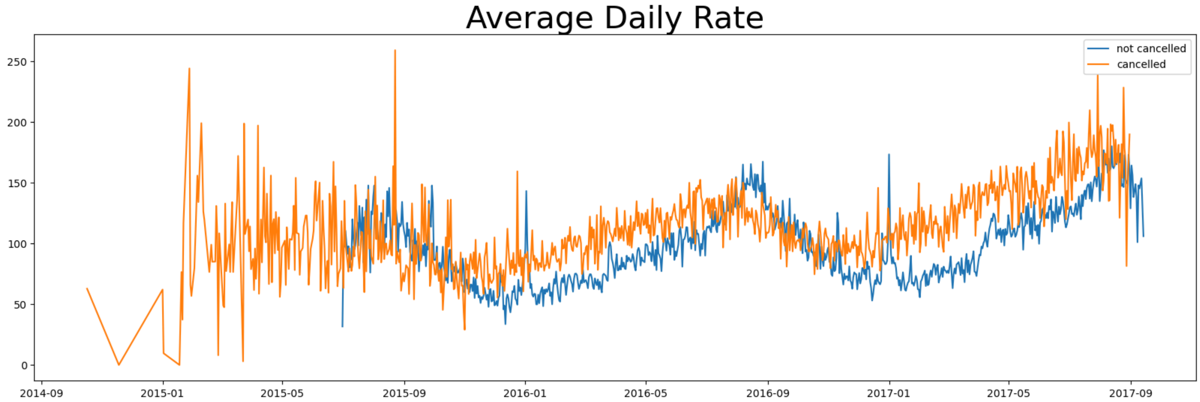


Insight:

There appears to be a positive correlation between higher ADR and cancellation rate. Possible reasons:

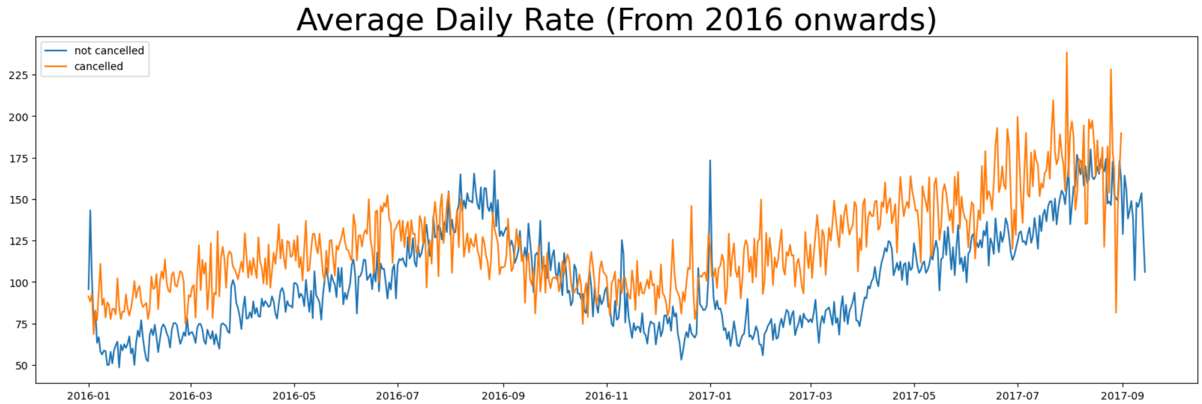
- Customers may cancel due to high price sensitivity.

- Promotions or overbooking strategies could be misaligned with customer expectations.



Visualization

The graph below illustrates ADR trends for canceled vs. non-canceled bookings (2016–2017):



Recommendations

1. **Price Optimization:**
    - Analyze pricing strategies for higher ADR dates to reduce cancellation risk.
2. **Cancellation Policy Review:**

- Implement flexible policies for higher ADR bookings.

3. **Customer Segmentation:**
   - Identify segments more likely to cancel high-value bookings and personalize offers.

Conclusion

The ADR analysis from 2016 onwards shows that cancellation rates tend to be higher with increased daily rates. Optimizing pricing and understanding customer behavior can help reduce churn and increase revenue.

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